$\cancel{\ln L = -n\ln\theta + \ln\prod x_i} \quad \cancel{L = \left(\frac{1}{\theta}\right)^n \mathbb{1}\{\min x_i \le \theta\}} \quad \frac{\max x_i}{2} \le \theta \le$

$\cancel{\ln L' = -\frac{n}{\theta}}$

[5] $\xi \sim R[\theta, 2\theta] \Rightarrow p(x) = \frac{1}{\theta}\{[\theta, 2\theta]\}$

$\vec{x}_n$ – выборка

a) $\alpha_1 = \int_{-\infty}^{\infty} x p(x)\,dx = \int_{\theta}^{2\theta} \frac{x}{\theta}\,dx = \frac{4\theta^2}{2\theta} - \frac{\theta^2}{2\theta} = \frac{3}{2}\theta = \bar{x}$

$\Rightarrow \tilde{\theta}_1 = \frac{2}{3}\bar{x}$ – ОММ

$L(\theta) = \left(\frac{1}{\theta}\right)^n \begin{cases} x_i \ge \theta \\ x_i \le 2\theta \end{cases} \forall i = \left(\frac{1}{\theta}\right)^n \begin{cases} \frac{\max x_i}{2} \le \theta \le \min x_i \end{cases}$

$= \left(\frac{1}{\theta}\right)^n \{\theta \in [\frac{x_{(n)}}{2}, x_{(1)}]$

$L'(\theta) = -n\theta^{n-1} < 0 \quad \forall \theta > 0 \Rightarrow$ max достигается в левой границе $\Rightarrow \hat{\theta}_2 = \frac{x_{(n)}}{2}$ – ОМП

б) Проверяем $\tilde{\theta}_1$: $M\tilde{\theta}_1 = M[\frac{2}{3}\bar{x}] = \frac{2}{3}\cdot\frac{1}{n}\sum_{i=1}^{n} Mx_i =$

$= \frac{2}{3}\cdot\frac{1}{n}\cdot n\cdot\frac{3}{2}\theta = \theta \Rightarrow \tilde{\theta}_1$ несмещенная

$D[\tilde{\theta}_1] = D[\frac{2}{3}\bar{x}] = \frac{4}{9}\cdot\frac{1}{n^2}\sum_{i=1}^{n} Dx_i = \frac{4}{9}\cdot\frac{1}{n}\cdot\frac{\theta^2}{12} =$

$= \frac{\theta^2}{27n} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_1$ - состоятельная

Проверяем $\tilde{\theta}_2$: $M\tilde{\theta}_2 = M[\frac{x_{(n)}}{2}] = \frac{1}{2}Mx_{(n)} =$

$= \frac{1}{2}F(x) = \cancel{\int_{-\infty}^{\infty}\frac{dx}{\theta}} = \frac{x-\theta}{\theta} \Rightarrow f_{(n)} \sim C_n^{\frac{x-\theta}{\theta}}$

$p(x) = n C_{n-1}^{\frac{x-\theta}{\theta}}\cdot\frac{1}{\theta} = \frac{n}{\theta}C_{n-1}^{\frac{x-\theta}{\theta}}$

$Mx_{(n)} = \int_{\theta}^{2\theta}\frac{n}{\theta}C_{n-1}^{\frac{x-\theta}{\theta}}x\,dx = \left\langle \begin{matrix} u = \frac{x}{\theta} & x = \theta u \\ du = \frac{1}{\theta}dx, & dx = \theta du \end{matrix}\right\rangle$

$= \int_{\theta}^{2\theta}\frac{n}{\theta}(u-1)^{n-1}\theta u\cdot\theta du = \theta\int_{\theta}^{2\theta} nu(u-1)^{n-1}du =$

$= \theta u(u-1)^n\Big|_{\theta}^{2\theta} - \theta\int_{\theta}^{2\theta}(u-1)^n du = \theta\cdot 2\theta(2\theta-1)^n - \theta^2(\theta-1)^n -$

$- \frac{\theta\cdot 2\theta(2\theta-1)^{n+1} - \theta^2(\theta-1)^{n+1}}{n+1} =$

$= \frac{2\theta^2(2\theta-1)^n(n+1) - \theta^2(\theta-1)^n(n+1) - 2\theta^2(2\theta-1)^{n+1} + \theta^2(\theta-1)^{n+1}}{n+1} =$

$= \frac{2\theta^2(2\theta-1)^n(n+1-2\theta+1) + \theta^2(\theta-1)^n(n+1-\theta-1-n-1)}{n+1} =$

$= 2\theta^2(2\theta-1)^n$

$= \int_1^2\frac{n}{\theta}(u-1)^{n-1}\theta u\theta du = \theta u(u-1)^n\Big|_1^2 - \theta\int_1^2(u-1)^n du =$

$= 2\theta + 2\theta - \theta\cdot\frac{1}{n+1}$

$\Rightarrow M\tilde{\theta}_2 = \frac{1}{2}Mx_{(n)} = \theta - \frac{\theta}{2n+2} = \frac{2n\theta+\theta}{2n+2} = \frac{\theta(2n+1)}{2n+2} \neq \theta$

$\Rightarrow \tilde{\theta}_2$ не явл. несмещенной

Исправляем $\tilde{\theta}_2' = \frac{2n+2}{2n+1} \cdot \frac{x_{(n)}}{2} = \frac{n+1}{2n+1} x_{(n)}$ - несмещенная

$$D\tilde{\theta}_2' = \left(\frac{n+1}{2n+1}\right)^2 D x_{(n)}$$

$$D x_{(n)} = M x_{(n)}^2 = \int_\theta^{2\theta} \frac{n}{\theta}\left(\frac{x-\theta}{\theta}\right)^{n-1} x^2 dx = \left\langle \begin{array}{l} u = \frac{x}{\theta} \\ du = \frac{1}{\theta} dx \end{array} \right\rangle$$

$$= \int_1^2 \frac{n}{\theta}(u-1)^{n-1} \theta^2 u^2 \cdot \theta dx = \theta^2 \int_1^2 n u^2 (u-1)^{n-1} du =$$

$$= \theta^2 \left(u^2 (u-1)^n \Big|_1^2 - \int_1^2 2u(u-1)^n du \right) = \theta^2 \left(4 - 3 - 2\frac{u(u-1)^{n+1}}{n+1} \Big|_1^2 + \right.$$

$$\left. + 2\int_1^2 \frac{(u-1)^{n+1}}{n+1} du \right) = \theta^2 \left(3 - \frac{2}{n+1} + \frac{(u-1)^{n+2}}{(n+1)(n+2)} \Big|_1^2 \right) =$$

$$= \theta^2 \left(3 - \frac{2}{n+1} + \frac{2+2}{(n+1)(n+2)} \right) = \cancel{\theta^2} \cdot \frac{3n^2 + \cancel{9n} + \cancel{6} - 2n - 4 + 2}{\cancel{(n+1)}(n+2)} =$$

$$= \theta^2 \cdot \frac{4n^2 + 12n + 8 - 4n - 8 + 2}{(n+1)(n+2)} = \theta^2 \cdot \frac{4n^2 + 8n + 2}{(n+1)(n+2)}$$

$$D x_{(n)} = \frac{4n^2 + 8n + 2}{(n+1)(n+2)} \theta^2 - \left(\frac{2n+1}{n+1} \theta \right)^2 =$$

$$= \theta^2 \left(\frac{(4n^2 + 8n + 2)(n+1) - (2 \cdot 4n^2 + 4n + 1)(n+2)}{(n+1)^2 (n+2)} \right) =$$

$$= \theta^2 \left(\frac{4n^3 + 4n^2 + 8n^2 + 8n + 2n + 2 - 4n^3 - 8n^2 - 4n^2 - 8n - n - 2}{(n+1)^2 (n+2)} \right) =$$

$$= \theta^2 \cdot \frac{n}{(n+1)^2 (n+2)} \Rightarrow D\tilde{\theta}_2' = \frac{(n+1)^2}{(2n+1)^2} \cdot \frac{\theta^2 n}{(n+1)^2 (n+2)} =$$

$$= \frac{\theta^2 n}{(2n+1)^2 (n+2)} \xrightarrow{n \to \infty} 0 \Rightarrow \tilde{\theta}_2' \text{ - состоятельная}$$

в) $D\hat{\theta}_1 = \frac{\theta^2}{2n}$ и $D\tilde{\theta}_2' = \frac{\theta^2 n}{(2n+1)^2 (n+2)}$

$$\frac{\theta^2}{2n} \geq \frac{n\theta^2}{(2n+1)^2 (n+2)} \quad \forall n \geq 3 \Rightarrow \tilde{\theta}_2' \text{ эффективнее } \hat{\theta}_1$$

2) $\exists\ f(x,\theta) = \frac{x_{(n)}}{\theta}$; Для $t \in [1,2]$: $F(t) = (t-1)^n$

$f(t) = n(t-1)^{n-1}$; $t_{\frac{1-\beta}{2}}$

Тогда $P(pt_{\frac{1-\beta}{2}} < \frac{x_{(n)}}{\theta} < pt_{\frac{1+\beta}{2}}) = \beta$

$\cancel{t_{\frac{1-\beta}{2}} = p_{\frac{1-\beta}{2}} = F(\frac{1-\beta}{2}) = \frac{1-\beta}{2}}$

$\Rightarrow p_{\frac{1-\beta}{2}} : F(t) = \frac{1-\beta}{2} = (t-1)^n \Rightarrow t = \sqrt[n]{(\frac{1-\beta}{2})} + 1$

$p_{\frac{1+\beta}{2}} = \sqrt[n]{(\frac{1+\beta}{2})} + 1$

$\Rightarrow P(\sqrt[n]{\frac{1-\beta}{2}} + 1 < \frac{x_{(n)}}{\theta} < \sqrt[n]{\frac{1+\beta}{2}} + 1) = \beta$

$P(\frac{x_{(n)}}{\sqrt[n]{\frac{1+\beta}{2}} + 1} < \theta < \frac{x_{(n)}}{\sqrt[n]{\frac{1-\beta}{2}} + 1}) = \beta$ - точный довер. интервал

e) Из ОММ $\tilde{\theta} = \frac{2}{3}\bar{x} = g(\tilde{\alpha}_1)$

$g(\alpha_1) = \theta$; $\nabla g = \frac{2}{3}$; $K = \alpha_2 - \alpha_1^2$

$\frac{\sqrt{n}(\tilde{\theta} - \theta)}{\sqrt{\nabla g(\tilde{\alpha}_2 - \tilde{\alpha}_1^2)}} = \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} \rightsquigarrow N(0,1)$

Для $N(0,1)$ $t_{1,2} = \mp 1{,}96 \Rightarrow P(t_1 < \frac{3n(\tilde{\theta} - \theta)}{2\sqrt{S^2(n-1)}} < t_2) = \beta$

$\Rightarrow P(\frac{2}{3}\bar{x} - \frac{3{,}92\sqrt{S^2(n-1)}}{3n} < \theta < \frac{2}{3}\bar{x} + \frac{3{,}92\sqrt{S^2(n-1)}}{3n}) = \beta$

- асимптотический доверительный интервал